

TAIR — Rubik Pi 3

Development Environment Setup Guide

Project	TAIR Indoor Mapping Drone
Board	Thundercomm Rubik Pi 3 (Qualcomm QCS6490)
OS	Ubuntu 24.04 LTS (Noble)
ROS Version	ROS 2 Jazzy
Setup Date	April 2026

Hardware Overview

Component	Details
SoC	Qualcomm Dragonwing QCS6490
CPU	8-core Kryo 670
RAM	8GB LPDDR4x
Storage	128GB UFS 2.2
NPU	Hexagon 770 — 12 TOPS AI performance
Connectivity	Wi-Fi 5, BT 5.2, USB 3.0, HDMI, 40-pin GPIO, CSI camera

Accessing the Board

SSH (Primary Method)

The board runs headless — always access it via SSH:

```
ssh ubuntu@<IP_ADDRESS>
```

Default credentials: Username: ubuntu | Password: ask Jashwanth

To find the board's IP on the network:

```
ping rubikpi.local
# Or check your router's connected devices for 'rubikpi'
```

■■ Both your laptop and the Rubik Pi must be on the same Wi-Fi network for SSH to work.

SSH is configured to auto-start on every boot. Verify with:

```
sudo systemctl is-enabled ssh    # Should print: enabled
sudo systemctl status ssh        # Should show: active (running)
```

Wi-Fi Management

```
# Scan available networks
sudo nmcli dev wifi list
```

```
# Connect with password
sudo nmcli dev wifi connect "NetworkName" password "YourPassword"
```

```
# Connect with no password
sudo nmcli dev wifi connect "NetworkName"
```

```
# Check new IP after connecting
ip addr show wlan0
```

■■■ After switching Wi-Fi, the IP address will change. Get the new IP before SSHing in.

ROS 2 Jazzy Setup

Why Jazzy (not Humble)?

	Humble	Jazzy
Ubuntu	22.04	24.04 ✓
Support until	2027	2029
Status on this board	Will NOT install	Installed ✓

ROS 2 is auto-sourced in ~/.bashrc on every terminal session. To verify:

```
ros2 pkg list | head -20
```

Workspace

The ROS 2 workspace for TAIR is at ~/tair_ws/

```
~/tair_ws/
■■■ src/      ← Put your ROS 2 packages here
■■■ build/
■■■ install/
■■■ log/

# Build the workspace
cd ~/tair_ws
colcon build
source install/setup.bash
```

Installed ROS 2 Packages

Package	Purpose
ros-jazzy-ros-base	Core ROS 2 runtime
ros-jazzy-slam-toolbox	LiDAR-based SLAM / mapping
ros-jazzy-pointcloud-to-laserscan	Convert 3D pointcloud to 2D scan
ros-jazzy-tf2-ros	Transform tree management
ros-jazzy-tf2-tools	TF debugging utilities
ros-jazzy-image-transport	Efficient camera data transport
ros-jazzy-image-pipeline	Camera processing pipeline
python3-colcon-common-extensions	Build tool for ROS 2 packages
python3-rosdep	Dependency management

Useful ROS 2 Commands

```
ros2 node list          # List all active nodes
ros2 topic list         # List all active topics
ros2 topic echo /topic_name # Monitor a topic in real time
ros2 topic hz /topic_name # Check topic publish rate
ros2 launch <package> <launch_file.py> # Launch a package
ros2 run <package> <node_name> # Run a single node
ros2 run tf2_tools view_frames # Check TF tree
```

VS Code Remote SSH

1. Install **VS Code** on your laptop
2. Install the **Remote - SSH** extension
3. Press **Ctrl+Shift+P** → type **Remote-SSH: Connect to Host**
4. Enter: **ubuntu@<IP_ADDRESS>**

You'll get a full VS Code editor connected directly to the board — file explorer, terminal, everything.

Rebooting / Shutting Down

```
sudo reboot    # Reboot
sudo poweroff  # Shutdown
```

Full Reset (Reflash OS)

■■ This wipes everything. Only do this if the board is seriously broken.

1. Download **Qualcomm Launcher** from Thundercomm's website
2. Download the **Ubuntu 24.04 image** for Rubik Pi 3
3. Put board into **EDL mode**: hold EDL button → plug in power → plug USB-C to PC
4. Open Qualcomm Launcher → select Rubik Pi + Ubuntu → **Flash**